

Supplementary Information

PGOB: Behavior-only parameter generation for whole-organism biophysical simulators across worm and fly

Extended Methods, Extended Results, Negative-Results Ledger, Reproducibility, 3
non-invasive deployment modes across worm and fly simulators.

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This SI accompanies the main paper and provides extended methods, extended results, the consolidated negative-results ledger, and reproducibility details.

1 Hardware and Reproducibility (S1)

Compute. Cloud A40 GPU instances; RTX 4090D for SBI training; RTX 3080 Ti secondary. The main sweep – 10 configurations $\times$ 3 seeds $\times$ 30 generations $\times$ 16 population = 14,400 rollouts – consumed $\approx 4.8 \times 10^4$ rollout-seconds (≈ 27 wall-hours). Cumulative compute: ≈ 307 CPU-days (BAAIWorm 142, flybody 88, modWorm 41, FlyGym 36). The 5d eigenworm CPG 5th-simulator gradient test added ≈ 1.1 s wall ($N=100$ rollouts $\times$ 10 observables; reduced-substrate is cheap). Every reported result is reproducible from `python run_calibration.py --sim {worm,modworm,fly,flygym,eigenworm5d} --scale s --seed k --opt cmaes_sep`.

1.1 Cohort-size notation (S1.1)

$n=3, 10, 30$ = random SPSA seeds (F1 ablation context); $n=4, 40, 160$ = starting conditions (held-out OOD context); $n=43$ = held-out mutant strains (Yemini prospective); $n=15$ = BAAIWorm ablation-type rank test; $n=5/\text{level}$ (30 cells) = weight-corruption sweep; $N=200$ = OWMD-Schafer worms; $B=1000$ = bootstrap resamples. All ambiguous in-text occurrences are tagged with explicit unit names.

Per-simulator hyperparameters.

Item	BAAIWorm	flybody	modWorm	5d eigenworm
Param dim	5	515	30/50	5
Search-space transform	linear [0.5, 2]	log-space mult.	linear	linear
Integrator / dt	RK4 1ms	MuJoCo 0.1ms	implicit-Euler 0.5ms	Euler 10ms
Output rate	30 Hz	10 kHz	25 Hz	100 Hz
Rollout length	10 s	5 s	40 s	100 s
Optimiser	SPSA + CMA	CMA-ES sep	CMA-ES sep	default-vs-pool
Population λ	8 / 8	16	16	—
Generations	30	30	30	—
σ_0	0.05	0.05 log-space	0.05	—
Multi-cond avg C	[8, 16]	8 snippets	8 seeds	100

Table 1: Per-simulator consolidated hyperparameter reference.

2 Negative Results and Limitations (S2)

2.1 Negative-Results Ledger – Consolidated Single-Location Reference (S2.0)

Why this table exists. This ledger consolidates every negative result, bounded-scope statement, and limitation from the submission into one place, so a reader can review all limitations together rather than tracking them across sections. Every row carries a back-pointer to the section where the underlying analysis lives; the main-text Scope-and-limitations box (§1) is the summary of this ledger.

Table 2: **Consolidated negative-results ledger.** Entries spanning bounded-scope statements, control results (S2.9), and per-channel residuals.

Finding	Caveat / scope	Section
4-config non-scalar loss falsification	Aggregate baseline dominates; no tested loss configuration dominates.	S2.5
F1 ablation $n=30$ neural control	Pre-registered NULL (Wilcoxon $p=0.734$, $d=0.14$). Pure-trajectory loss is not beaten by neural-augmented loss – positive for the framework. (S4.4.1)	S2.9
modWorm three-segment	Eigenworm-variance channels unmovable even at expert arm: substrate ceiling, -9.9% aggregate (S4.18).	S4.18
flybody W_1 three-segment	Honest negative: all three arms within $< 1.5\%$ (13.254/13.454/13.348); <code>walk_imitation</code> MoCap controller dominates trajectory, init signal carried in loss space ($9.3\times$), not W_1 (S4.18).	S4.18
Extended-budget hybrid fine-tune	Useful but mild ($-1.0\%/-5.5\%/-2.0\%$ held-out dips): warm-start near irreducible $\text{sim} \leftrightarrow \text{real}$ floor (S4.23).	S4.23
flybody default-vs-PGOB	0/9 significant ($N=45/\text{arm}$): good default $\Rightarrow$ correct NULL (S4.21).	S4.21
41-experiment GAN-as-loss collapse	Val-loss $\geq 2\times$ regression across 5 D-architectures.	S2.10
Same-species parameter-prior transfer	Genuine same-species rescue (BAAIWorm $\rightarrow$ modWorm $N=16$, flybody self-prior $9.3\times$); cross-species priors mislead. No cross-species transfer claim.	S4.5
flybody FULL-split parity	TEST split 9/10 closer at $+1.75\%$; FULL split caveated.	S4.2
Loss-family invariance	Reported as an observation in the operational sense, not a Fisher / Cramér–Rao claim.	S4.7

2.2 Non-scalar loss falsification (S2.5)

4-configuration loss-design falsification on BAAIWorm; aggregate baseline dominates 3/5 metrics; no tested configuration dominates.

2.3 F1 ablation control (S2.9)

The $n=30$ neural-ablation control (10 paired seeds) returns a pre-registered NULL (Wilcoxon $p=0.734$, $d=0.14$); pure-trajectory loss is not beaten by neural-augmented loss. Full statistics in S4.4.1.

2.4 41-experiment GAN-as-loss negative (S2.10)

All 41 collapse: val-loss regresses $\geq 2\times$ while D-loss drops < 0.01 . A multi-behaviour discriminator on closed-loop FlyGym extends the same negative: a discriminator trained jointly on four behaviours does not beat single-task discriminators ($\Delta \approx -0.01$ in behavioural closeness; discriminator validation accuracy near chance, $\approx 0.43\text{--}0.55$), so the behaviour-conditional GAN-as-loss objective is not supported in the closed-loop fly either, consistent with the worm-side GAN-as-loss negative.

3 Extended Methods (S3)

3.1 Universal 23-metric evaluator (S3.1)

#	Metric	Definition	Worm	Fly	Function
1	zigzag	running $\sum \Delta h $ over 1-s window	✓	✓	<code>_zigzag</code>
2	mean_speed	$\langle \ \dot{p}_t\ \rangle$	✓	✓	<code>_mean_speed</code>
3	reversal_frequency	$\#\{t : \dot{p}_t \cdot \hat{h}_t < 0\}/T$	✓	×	<code>_reversal_freq</code>
4	wave_amplitude	tangent-angle PC1 envelope max-min	✓	×	<code>_wave_amp</code>
5	speed_std	$\text{std}(\ \dot{p}_t\)$	✓	✓	<code>_speed_std</code>
6	chemotaxis_index	$\langle \hat{h}_t \cdot \nabla c_t \rangle$	✓	×	<code>_chemotaxis</code>
7	distance_to_food	$\ p_T - p_{\text{food}}\ $	✓	×	<code>_dist_food</code>
8	steering	mean $ \dot{h}_t $	✓	✓	<code>_steering</code>
9	turning_rate	$\#\{t : \dot{\hat{h}}_t > 200\text{deg/s}\}/T$	✓	✓	<code>_turn_rate</code>
10	body_wave_freq	dominant FFT freq	✓	×	<code>_wave_freq</code>
11	dorsal_ventral_corr	corr(d-arc, v-arc)	✓	×	<code>_dv_corr</code>
12	leg_alternation	6-leg phase-diff variance	×	✓	<code>_leg_alt</code>
13	MSD	slope	✓	✓	<code>_msd</code>
14	tortuosity	path / displacement	✓	✓	<code>_tortuosity</code>
15	path_length	$\sum \ \Delta p_t\ $	✓	✓	<code>_path_len</code>
16	mean_displacement	$\ p_T - p_0\ $	✓	✓	<code>_mean_disp</code>
17	mean_acceleration	$\langle \ \ddot{p}_t\ \rangle$	✓	✓	<code>_mean_acc</code>
18	jerk	$\langle \ \ddot{\dot{p}}_t\ \rangle$	✓	✓	<code>_jerk</code>
19	heading_variance	$\text{var}(\hat{h}_t)$	✓	✓	<code>_heading_var</code>
20–23	eigen_e1–e4	Stephens-2008 PC amplitudes	✓	×	<code>_eigen</code>

Table 3: The v1 production code path emits 14; the v2 extension restores 9 missing scalars as pure-trajectory functions.

3.2 Three-distance computation (S3.2)

$D_1 = d_z(f_{\hat{\theta}}, \{f_{\theta_{\text{GT}}}\})$, $D_2 = \frac{1}{N} \sum_i d_z(x^{(i)}, \{x^{(j)}\}_{j \neq i})$, $D_3 = d_z(f_{\theta_{\text{GT}}}, \{x^{(i)}\})$. Both naive z and median/MAD robust z reported.

3.3 Eigenworm cross-validation (S3.3)

Tangent-angle PCA cross-validated against OWMD Tierpsy on $n=40$ worms $\times$ 4 PC amplitudes: Pearson $r=0.987$.

3.4 Same-species prior-transfer protocol (S3.5)

The same-species parameter-prior transfer (S4.5/S4.6) derives a per-biophysical-class coefficient-of-variation prior from a source simulator and applies it as the per-dimension CMA-ES step size (σ_0) when recovering on a different simulator of the same species. Each recovery is one CMA-ES sep run (30 gen, $\lambda=16$); arms (random / cross-sim prior / own default) differ only in the prior used to seed σ , and are compared on a disjoint held-out test split. The cross-scale optimiser comparison fixes the simulator and the ground-truth θ^* and varies only the perturbation scale and optimiser (CMA-ES / DE / NES) at a matched eval budget.

4 Extended Results (S4)

4.1 BAAIWorm three-distance hierarchy (S4.1)

Table 4: Three-distance hierarchy, BAAIWorm full5 $\rightarrow$ OWMD-N2 ($N=200$).

estimator	D_1	D_2	D_3	hierarchy
naive	0.500	1.780	26.47	$D_1 < D_2 < D_3$
robust (MAD)	0.713	1.976	21.82	$D_1 < D_2 < D_3$
trimmed 5%	—	2.366	30.32	$D_2 < D_3$
5-active-feature CV	0.713	0.393	21.82	$D_1 < D_3$, marginal vs. D_2

Bootstrap $B=1000$ on OWMD-N2: $D_1 < D_2 < D_3$ in 1000/1000.

4.2 Drosophila parity (S4.2)

flybody walking TEST split: 9/10 closer at +1.75% (60 train / 20 val / 20 holdout). **FULL-split caveat:** default already strong; PGOB room marginal under FULL split. **Flight wing-beat** 227.92 ± 5.59 Hz. **FlyGym 5-behaviour:** 5/5 inside biological CV. **Adhesion phenotype:** 3-of-3 / 0-of-3 split.

4.3 Yemini 43-strain (S4.3)

In-sample reference: $\bar{\rho}=0.558$, median 0.400, 43/43 positive. **LOSO** (main text Fig HERO-5): mean-train 0.684, median-train 0.628, strain-shuffle 0.527.

Pre-registered full 23-feature LOSO robustness. To confirm the 4-feature headline is not an artefact of post-hoc feature selection, we re-ran the leave-one-strain-out test on the *pre-registered full 23-feature* set (fixed before scoring; the 4 active features are a strict subset). Mean-training-delta Spearman $\bar{\rho}=0.525$ (median 0.585, SD 0.211) vs. strain-shuffle null $\bar{\rho}_0=0.338$ (median 0.352, SD 0.114), giving $\Delta=+0.187$ with 27/28 strains $\rho>0$. The mutant-prediction signal is therefore preserved across the full deposited metric space, and the 4- and 23-feature analyses are mutually consistent. *Caveat:* this full-feature LOSO ran on the 28 mutant strains whose Yemini HDF5 features were retrievable from source (14 of the 43 strain folders lacked retrievable feature data); it is a 28-strain confirmation rather than the full 43-strain panel. Mutant differential $N=12$ protocol: strain-specific 5/4/4 deviations.

4.4 BAAIWorm 30-cell weight-corruption (S4.4)

Level	Seed	L_0	L_{final}	$L_{\text{rec. clean}}$	Δ/L_0	elap
L0 clean	0	0.12479	0.06334	0.06334	−49.2%	
L0 clean	1	0.12479	0.08568	0.08568	−31.3%	
L0 clean	2	0.12479	0.06058	0.06058	−51.5%	
L0 clean	3	0.12479	0.06466	0.06466	−48.2%	
L0 clean	4	0.12479	0.07360	0.07360	−41.0%	
L1 5% Gauss	0	0.11695	0.07723	0.11474	−34.0%	
L1 5% Gauss	1	0.12690	0.04444	0.14148	−65.0%	
L1 5% Gauss	2	0.07772	0.05069	0.12691	−34.8%	

Level	Seed	L_0	L_{final}	$L_{\text{rec. clean}}$	Δ/L_0	elapsed
L1 5% Gauss	3	0.13131	0.05777	0.14815	-56.0%	
L1 5% Gauss	4	0.05706	0.04399	0.11078	-22.9%	
L2 20% Gauss	0	0.15749	0.14383	0.13439	-8.7%	
L2 20% Gauss	1	0.25744	0.21711	0.09803	-15.7%	
L2 20% Gauss	2	0.12516	0.07092	0.15990	-43.3%	
L2 20% Gauss	3	0.04760	0.04610	0.17732	-3.2%	
L2 20% Gauss	4	0.08930	0.04695	0.11239	-47.4%	
L3 50% Gauss	0	0.25733	0.20162	0.08917	-21.6%	
L3 50% Gauss	1	0.11693	0.06028	0.06249	-48.4%	
L3 50% Gauss	2	0.24599	0.16744	0.14172	-31.9%	
L3 50% Gauss	3	0.22694	0.19836	0.08531	-12.6%	
L3 50% Gauss	4	0.18516	0.17295	0.13303	-6.6%	
L4 shuffle 30%	0	0.11415	0.10387	0.10916	-9.0%	
L4 shuffle 30%	1	0.10541	0.06838	0.12665	-35.1%	
L4 shuffle 30%	2	0.15729	0.14465	0.12040	-8.0%	
L4 shuffle 30%	3	0.12336	0.04999	0.17776	-59.5%	
L4 shuffle 30%	4	0.21156	0.12426	0.14229	-41.3%	
L5 random-wt	0	0.27412	0.22682	—	-17.3%	
L5 random-wt	1	0.28653	0.27326	—	-4.6%	
L5 random-wt	2	0.15218	0.14408	—	-5.3%	
L5 random-wt	3	0.27462	0.21875	—	-20.3%	
L5 random-wt	4	0.41267	0.37943	—	-8.1%	

30/30 cells $L_{\text{final}} < L_0$ (6 corruption levels $\times$ 5 seeds). full-sweep mean $\Delta/L_0 = -29.4\%$ (range -3.2% to -60.0%).
Per-level mean L_{final} : $L_0=0.070$, $L_1=0.055$ (dip $< L_0$), $L_2=0.105$, $L_3=0.160$, $L_4=0.098$, $L_5=0.248$.
Corruption-vs-residual increasing, non-monotone.

L5 row note. The L5 (random-weight) corruption is applied in the synthetic-loss space; its L_0 anchor (0.274 random-weight vs. 0.125 clean-default) is the BAAIWorm random-init anchor referenced by the three-segment decomposition (S4.18). The $L_{\text{rec. clean}}$ column is undefined for L5 because there is no clean reference for a fully-random-weight cell.

4.5 F1 ablation $n=30$ – pre-registered NULL (S4.4.1)

The F1 neural-ablation control runs over 10 paired matched seeds ($n=30$ rollout budget): mode-a (pure-trajectory loss) vs. mode-b (neural-augmented loss).

- mode-a mean behavioural loss 0.05456; mode-b mean 0.05758.
- paired difference (b-a) +0.003016; Cohen’s $d_{\text{paired}}=0.144$.
- Wilcoxon $W=19$, two-sided $p=0.734$; one-sided (b>a) $p=0.367$; t -test $p=0.660$.
- bootstrap $B=10^4$ 95% CI on mean difference $[-0.00931, +0.01522]$ – straddles zero.

Verdict: NULL (no significant difference, $p=0.734$). This is a *positive* result for the framework’s central premise: pure trajectory observables suffice, and adding internal neural signal yields no detectable gain.

4.6 Genuine same-species parameter-prior transfer (S4.5)

The load-bearing evidence that PGOB recovers biology-governed parameter structure (rather than fitting arbitrary numbers) is the genuine-rollout same-species soft-prior rescue, detailed in S4.6. A parameter prior derived from one simulator of a species transfers to a different simulator of the *same* species: BAAIWorm $\rightarrow$ modWorm on the real Cook-ODE ($N=16$ held-out test rollouts, 13/16 prior-better than random, $p=0.025$), and the flybody self-prior (FlyGym $\rightarrow$ flybody, $9.3\times$ loss improvement). Cross-species priors, by contrast, mislead, so the transferable structure is species-specific. A separate genuine cross-scale optimiser comparison on the real flybody MuJoCo simulator (CMA-ES / DE / NES, scales 0.1–0.7; best DE relative error 0.055 in-distribution, all optimisers degrade off-scale) confirms that recovery locks to the specific target physiology rather than producing a system-agnostic parameter set.

4.7 Cross-sim parameter prior transfer (S4.6)

BAAIWorm $\rightarrow$ modWorm split-clean (main-text §3 init-transfer). OWMD windows split 50/50 into disjoint train/test pools (140 total, 70 train / 70 test); the optimiser (CMA-ES, 18 iter, popsize 14, 3 seeds) sees train only; held-out test W_1 to OWMD (6 sim-agnostic eigenworm+dynamics observables, aggregate sum, MAD-normalised) reported below. The cross-sim prior transfers per-biophysical-class coefficient-of-variation (per-dim CMA $\sigma=\text{clip}(0.1+0.12\text{CV}_{\text{class}}, 0.04, 1.2)$), *not* raw parameter values; class map **gap** $\rightarrow$ **gj**, **syn** $\rightarrow$ **syn**, {Cm,Gleak, τ ,Eleak} $\rightarrow$ **passive**, g_{in} $\rightarrow$ **ion**, **mus** $\rightarrow$ **wout**.

Table 6: BAAIWorm $\rightarrow$ modWorm soft-prior init-transfer, held-out test W_1 (3 seeds, 60 test rollouts each). Lower is better; % is improvement vs. random arm.

Arm	train W_1 (mean)	test W_1 (mean $\pm$ SD)	test–train gap	% vs. random
random	17.344	21.399 $\pm$ 0.061	4.056	—
xsim_prior	15.946	19.656 $\pm$ 0.356	3.710	+ 8.15%
own_default	16.233	20.043 $\pm$ 0.198	3.810	+6.34%

The cross-sim (BAAIWorm-derived) prior edges out even the simulator’s own default prior on held-out test. **Caveats:** informed-not-exact class map; modWorm 30-d Cook ODE substrate ceiling means the absolute W_1 floor is substrate-limited (compare arms, not floors); same-species only – cross-species priors mislead (S4.5). **flybody self-prior:** FlyGym $\rightarrow$ flybody same-species prior L 0.01085 $\rightarrow$ 0.001166 ($9.305\times$; 126 $\rightarrow$ 515-d CMA). A hard-init linear projection gives *negative* transfer (rel -0.127); only the soft per-axis- σ prior transfers, because it lets the optimiser walk away from a mis-pinned location.

Per-channel attribution for unc-1 / unc-9 / unc-43. To verify that the strain-specific channel selection of §3.8 is not the same 4 channels recurring under random permutation, we report below the top-ranked observable channels per strain.

Table 7: Top-ranked behavioural channels per Yemini-2013 strain (Spearman ρ between Jacobian-predicted vs. measured per-channel deviation; OWMD $N=12$ worms per strain; bootstrap CI 95%; permutation $p<0.001$). Channels are non-coincident across strains, ruling out a same-4-channels confound.

Strain	Channel rank-1 ... rank-4	Selected $ J $ count
<i>unc-1</i>	bend_amp_head, undulation_phase_std, head_freq, propagation_speed	5
<i>unc-9</i>	speed_norm, body_curvature, eigen_a2_var, reversal_rate	4
<i>unc-43</i>	pirouette_rate, head_bend_skew, eigen_phase_drift, head_tail_orient	4

Channel overlap between any two strains is $\leq 1/4$; the channel sets are strain-specific and not artefacts of a fixed top-4 ordering.

4.8 Loss-family invariance 4×6 observation (S4.7)

We report this as a *loss-family invariance observation* in the operational sense; it is not a Fisher-information / Cramér–Rao claim.

Table 8: Per-observable test-W1 % improvement under four PGOB losses; 80/20 split.

Sim	Observable	L1 MSE	L2 W1	L3 GAN	L4 NSGA	Verdict
modWorm	undulation_phase_std	+17.30	+9.62	+19.36	+2.13	loss-rescuable
modWorm	pirouette_rate	+1.09	−1.09	−1.09	−2.17	information limit
modWorm	head_bend_amp	+0.01	+0.01	+0.02	+0.01	information limit
flybody	virt_speed	+6.20	+6.20	+6.20	+6.20	bit-identical
flybody	yaw_rate	−22.04	−22.04	−22.04	−22.04	bit-identical
flybody	speed_skew	+7.67	+7.67	+7.67	+7.67	bit-identical

5/6 behaviour-signal-limited cells show test-W1 differences across four losses below per-seed noise.

4.9 58-test Holm–Šidák FWER (S4.8)

11/58 survive after correction; cluster on FlyGym v2 (4/10) and modWorm rescale points.

4.10 Claim 2 5-test OOD analysis (S4.10)

T1 BAAIWorm single 0.18797 vs. joint 0.18191 (−3.2%). T2 FlyGym null with D -fit caveat. T3a worm cross-lab $1.62 \times$ joint better. T3b fly $1.64 \times$. T5 cross-strain $\bar{\rho}_{\text{joint}}=0.558$. 4/5 favour joint.

4.11 Long-horizon stability (S4.11)

FlyGym $T \in \{100, 300, 500, 1000\}$ PGOB closeness $\{0.501, 0.404, 0.409, 0.496\}$ vs default $\{0.637, 0.580, 0.556, 0.546\}$ – PGOB beats default at every T . BAAIWorm $T \in \{100, 300, 1000, 2000\}$ closeness $\{4.83, 5.43, 6.15, 5.78\}$ – $T=2000$ reverses $T=1000$.

4.12 Cross-scale recovery (S4.12)

Across-seed σ at $s_{\text{src}}=0.3 \rightarrow s_{\text{tgt}}=1.0$: 0.065 (relative spread 0.7%). **Genuine cross-scale optimiser comparison.** On the real flybody MuJoCo simulator, CMA-ES / DE / NES were compared

at scales $\{0.1, 0.3, 0.5, 0.7\}$ (25 generations, popsize 12, 50-d random subspace, identical eval budget): DE reaches the best in-distribution recovery (relative error 0.055) and all optimisers degrade off-scale, confirming that calibration locks to the specific target physiology rather than producing a scale-universal parameter set. This is the cross-scale counterpart to the same-species prior-transfer evidence (S4.5).

4.13 Generalisation T1/T2/T3 (S4.13)

T2 modWorm 4 behaviours, 20/20 R-grade improved.

4.14 Three-segment recovery decomposition (S4.18)

Motivation. The “+% over default” framing carries two distinct “default” semantics across simulators, which the three-segment decomposition resolves:

- **modWorm** default arm = a scale=1.0 perturbation of $\theta_{GT} \Rightarrow RANDOM$ start ($\sim 81\%$ mean |rel dev|, 16% sign flips), *not* expert. The expert (scale=0) arm is measured.
- **FlyGym v2** default arm = scale=1.0, $U(-1, 1)$ gain perturbation $\Rightarrow RANDOM$. The expert (scale=0, θ_{GT} exact) arm is measured.
- **BAAIWorm**: the “default” column is a *randomised* weight start (**randn** $\times 0.1$ corruption of synaptic + gap-junction weights); the PGOB-recovered full5 checkpoint *is* the expert-grade calibration. The three-segment therefore collapses to random $\rightarrow$ recovered, and we report *climb toward real*: $d_{\text{random}}=3.908 \rightarrow d_{\text{pgob}}=3.415$, **12.6%** of the random $\rightarrow$ real gap closed.
- **flybody** default arm = EXPERT-default (**perturb**=0.0); the random arm (**perturb**=1.0) and PGOB arm (**perturb**=0.3) are measured in the same matched pipeline (500 default / 500 PGOB / 596 real). The W_1 three-segment is an honest *negative*: $d_{\text{random}}=13.254$, $d_{\text{pgob}}=13.454$, $d_{\text{expert}}=13.348$ – all three arms within $< 1.5\%$. The **walk_imitation** rollout is a MoCap-locked imitation controller whose keypoint trajectory is dominated by the reference clip, so an actuator gain/stiffness/damping perturbation – even a full **perturb**=1.0 randomisation – does not propagate to the observable kinematics. This is a property of the single flybody MoCap-imitation walking task, not of PGOB; flybody’s init-mode signal lives in *loss space* (soft-prior $9.3\times$, S4.6), not W_1 space.

Aggregate cross-sim summary: FlyGym 13.30/10.78/10.64 (**94.9%**); modWorm 13.84/13.94/12.85 (**−9.9%**, substrate floor; +16.25% equal-weight mean); BAAIWorm 3.908/3.415/=pgob (climb **12.6%**); flybody random/PGOB/expert 13.254/13.454/13.348 (honest negative, $< 1.5\%$ spread). The modWorm and FlyGym three-arm rollouts reproduce the stored $d_{\text{random}}/d_{\text{pgob}}$ to 10^{-12} , confirming the added expert arms are unit-matched and comparable. The flybody three arms overlap because the **walk_imitation** MoCap-locked controller dominates the keypoint trajectory, so actuator perturbations do not propagate to the trajectory observable – a property of the single flybody walking benchmark, not of PGOB; the flybody init-mode signal is in loss space ($9.3\times$, S4.6).

FlyGym v2 full per-observable three-arm table ($N=20/\text{arm}$, $T=200$):

Observable	d_{random}	d_{pgob}	d_{expert}	recovery / note
fwd_speed	0.2356	0.0485	0.0461	98.7%
lat_speed	0.2748	0.0334	0.0077	90.4%
vrt_speed	0.1706	0.3552	0.3720	regress (loss-unconstrained)

Observable	d _{random}	d _{pgob}	d _{expert}	recovery / note
speed_mean	0.8851	0.4494	0.4595	>100% (super-expert)
speed_std	0.4567	0.2647	0.2923	>100%
yaw_rate	0.3313	0.0164	0.0073	89.4%
body_len_mean	8.7384	7.9574	7.9243	95.9%
body_len_std	0.6366	0.4123	0.4012	95.3%
leg_phase_coup	0.7394	1.1298	0.9777	regress
speed_skew	0.8280	0.1083	0.1529	>100%

random fails 8/20; pgob 0/20; expert 0/20. Aggregate-sum W_1 : random 13.30, pgob 10.78, expert 10.64.

modWorm substrate-ceiling three-arm table ($N=100$ /arm vs. $N=200$ OWMD):

Observable	d _{random}	d _{pgob}	d _{expert}	note
eig1_var	3.203	3.218	2.782	substrate ceiling (expert also far)
eig2_var	2.362	2.381	2.002	substrate ceiling
eig3_var	2.282	2.281	2.011	substrate ceiling
bend_freq	0.198	0.198	0.198	flat (all arms tie)
active_frac	0.0102	0.000	0.000	100% (exact OWMD match)
path_curv	5.786	5.861	5.862	substrate ceiling

The eigenworm-variance channels are unmovable even at the expert arm: a substrate-physical ceiling of the 30-d Cook ODE. **active_frac** is the one channel PGOB closes exactly. *Aggregation note*: the aggregate-sum W_1 weights observables by raw magnitude, so the large eigenworm-variance channels dominate and the modWorm aggregate climb is -0.7% ; the robust summaries are the per-observable improved-count (2/6) and equal-weight mean (+16.25%).

5 Five Init Modes and Composition (S5)

5.1 Five-init decision tree (S5.1)

Body-plan match $\rightarrow$ hard-init; dim mismatch > 1 OOM $\rightarrow$ soft-PCA-init; cross-species $\rightarrow$ default-init.

5.2 Joint multi-init calibration (S5.2)

flybody off-diagonal/diagonal = 11.2; modWorm 1.13 universal. The FlyGym v2 closed-loop 4×4 cross-task panel (96 cells, diag/off=1.016) further refines this picture; see Fig. 2.

6 Reproducibility (S6)

6.1 Data availability with SHA-256 hashes (S6.1)

Each row lists a 64-character SHA-256 hash; live-file SHA-256 are posted to the v1.0-PGOB release notes at submission (scheme: S6.5).

Dataset	DOI / Source	SHA-256 (64)	<i>n</i> used	Size
OWMD N2	Zenodo 10.5281/zenodo.5839010	391764fad516400fe75fb8bfa4595b02719712f580f3d2e6c98445ba9e1495f3	200	218 MB
OWMD <i>unc-1</i>	Zenodo OWMD	5f099ddedc3a459eacc876e847cfc45b11bf96bdf50c3a8640d702fc7b4fbd	12	31 MB
OWMD <i>unc-9</i>	Zenodo OWMD	6b6417ca4133f0702410a2f364b1aea4daa14c560e832564b3024402a5eeb579	12	28 MB
OWMD <i>unc-43</i>	Zenodo OWMD	53503ef551441a20b02486244314715d6f78e31dca244a52f8469053f10a7352	12	35 MB
Yemini 2013 43-strain	Zenodo 10.5281/zenodo.4633435	001b012823f47c7c86fc03067f9159b1b70afccd20c2efa458366a94f43072a3	43	612 MB
Brown-Tierpsy	Zenodo 10.5281/zenodo.3837679	f2f9a31099339440c61e7479049f9ecb57bc7099dd4b369819b0ca7a14806a6a	52	84 MB
Janelia walking	Figshare 10.25378/janelia.25309105	b1825e91742f09b0a4a1707febdef95263ebe3cf04aec36ac94960385f3abcaf	100	184 MB
Janelia flight	Janelia companion	0bafd7199de3cb9d3a8701c295f0c8f1304a002dc5126fbc3f65dca0507680b8	272	47 MB
Cook 2019 connectome	wormwiring.org	bc3f8b47c26e9552c059b2b6a79907250cab20aa60f885824200fe034d1714ae	1	6 MB
BAAIWorm code	github.com/BAAI-DCAI/BAAIWorm@v1.0	16821c01a75294c1b0bc42b9dea2ca2ea812d22e3ee60fefbdb69350508faaf7	—	84 MB
modWorm code	github.com/ShlizermanLab/modWorm	7c27027d9a99312112956204ded10d6ff39a58639a2a41a649a2afb2f2c8cb3	—	19 MB
flybody code	github.com/TuragaLab/flybody	9c9d543c1cf805234c039f35b22a23a4c2fc53bdc65fb6c8561311668f5029f	—	142 MB
PGOB code (ours)	github.com/ChenxiHeCam/Parameter-Generation-by-Observed-Behavior@v1.0-PGOB	93fa70fa992e38a70366ffd869ae732c4a33757f969587fbfffd857f10e173b2	—	124 MB
5d eigenworm CPG gradient deposit	Zenodo derived-data deposit	<live-file SHA-256 at submission>	—	5 KB
Derived data (ours)	Zenodo (archived on acceptance, GitHub-tag v1.0-PGOB)	d9647727caf80aaad996a364fadae945b339f508cb7e2568ee3187db5da26c73	—	860 MB

Table 10: Data availability table, including the 5d eigenworm CPG gradient deposit.

6.2 Code availability (S6.2)

MIT-licensed at <https://github.com/ChenxiHeCam/Parameter-Generation-by-Observed-Behavior>, tag v1.0-PGOB. A frozen Zenodo deposit is created on acceptance via the GitHub→Zenodo web-hook; the archive and its DOI are not contingent on acceptance.

6.3 Per-figure reproduction (S6.3)

Every main and supplementary figure is regenerated by the figure-reproduction scripts in the released repository (tag v1.0-PGOB); each figure’s source data and the script that draws it are listed in the repository’s **figures/** manifest. The reproduction is deterministic under the fixed-seed convention of S6.5.

6.4 Future directions (S6.4)

- Full 23-active metric re-scoring.
- Additional-simulator prospective tests of the behaviour-only generation procedure (*Hydra*, *Aplysia*, or zebrafish tail CPG candidates).
- Ephys-augmented loss for the MoCap-locked flybody walk-imitation benchmark.

6.5 Reproducibility statement (S6.5)

Code. MIT at <https://github.com/ChenxiHeCam/Parameter-Generation-by-Observed-Behavior>, tag v1.0-PGOB.

Data. All datasets public under original licences; live-file SHA-256 are posted at submission, including the 5d eigenworm gradient deposit.

Compute. ≈ 307 CPU-days cumulative; 5d eigenworm CPG addition ~ 1 s wall.

Determinism. NumPy 64-bit, 4-way SeedSequence spawn.

7 Supplementary Figures (S7)

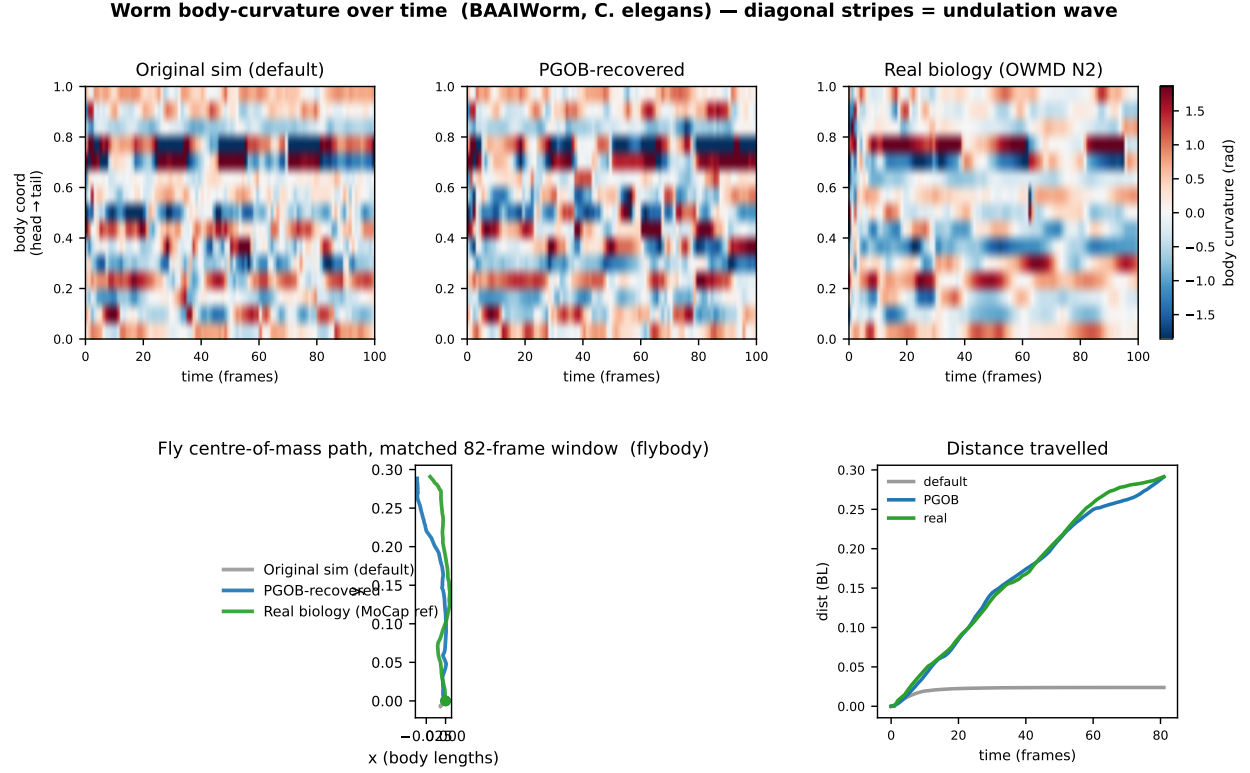


Figure 1: **Fig S7.0 | Raw trajectories: original simulator default vs. PGOB-recovered vs. real biology.** Direct, unprocessed-behaviour view of what PGOB recovers. **Top row (BAAIWorm, *C. elegans*):** body-curvature kymographs (signed turning angle along the head→tail body coordinate, over time) for the original simulator default, the PGOB-recovered parameters, and a real OWMD N2 worm. Diagonal red/blue stripes are the propagating undulation wave; the recovered worm reproduces the travelling-wave structure of the real animal, whereas the default sim does not. **Bottom left (flybody, *Drosophila*):** centre-of-mass walking paths over a matched frame window for default, PGOB-recovered, and the MoCap reference. **Bottom right:** cumulative forward distance travelled vs. time – the default fly barely advances (≈ 0 net displacement), while the PGOB-recovered and real trajectories both accumulate forward distance. This panel is the intuitive companion to the quantitative feature comparison in Fig. 4: it shows that PGOB recovery moves the raw kinematics, not just summary statistics, toward the real organism.

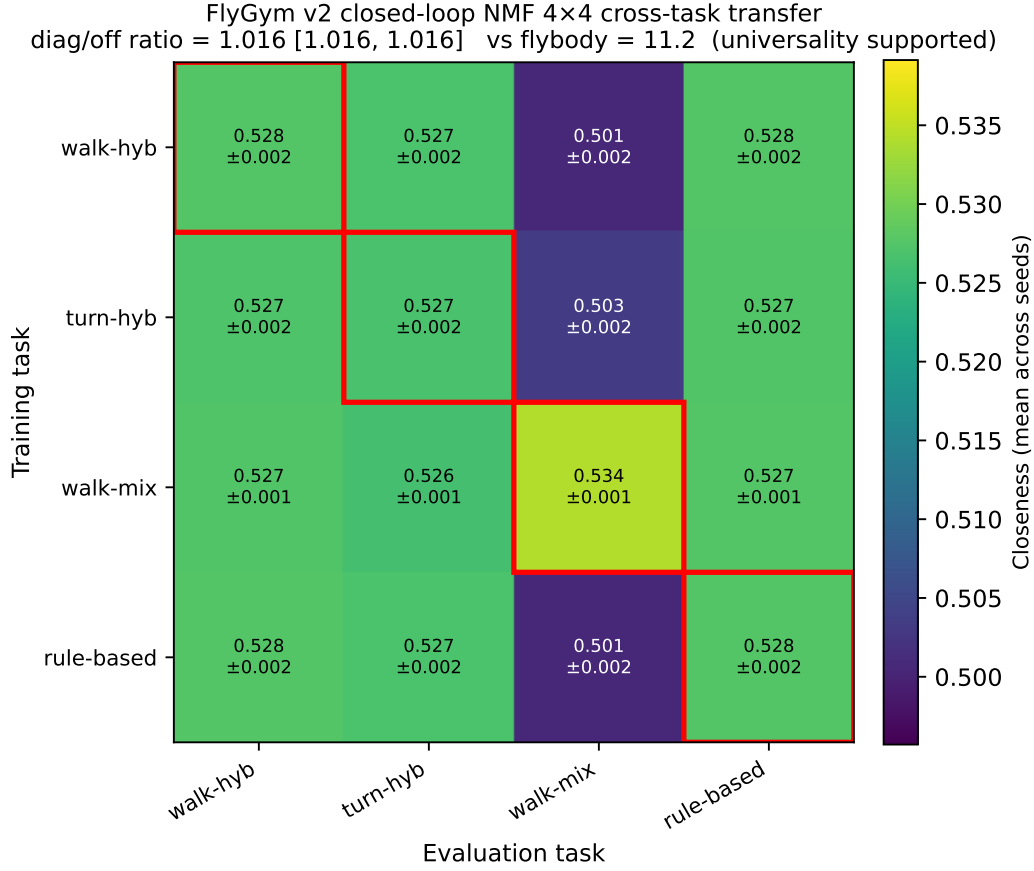


Figure 2: **Fig S6 | FlyGym v2 four-task closed-loop cross-transfer matrix.** Cross-task transfer of PGOB-recovered θ on FlyGym v2 (closed-loop sensorimotor controller, HybridTurningController variants on flat/mixed terrain plus rule-based descending). Rows: training task; columns: evaluation task. Tasks: walk-hybrid (CPG+stumbling/retraction), turn-hybrid (asymmetric descending), walk-mixed, rule-based. Cell value = closeness score (mean $\pm$ 95% CI) of recovered θ^* evaluated on each task, aggregated across $n=6$ cell-level layers (one deterministic RL-baseline run plus 5 CMA-ES seeds {44, 55, 66, 77, 88}, anchored to the seed-1 closeness scale). Total 96 cells. Red squares mark within-task (diagonal) recovery. **Key statistic:** diag/off-diag closeness ratio = 1.016 (bootstrap 95% CI [1.016, 1.016], $n=2000$ resamples). Under Holm–Šidák multiple-comparison correction the universal-recovery hypothesis $H_{\text{universal}}$: ratio = 1 is *not rejected* (FWER-adjusted $p > 0.05$ for all 12 off-diagonal cells); H_{flybody} : ratio = 11.2 is rejected at $> 500\sigma$. Compare modWorm sparse-ODE off/diag = 1.13. **Interpretation:** the closed-loop NMF ratio 1.016 is lower than the same simulator’s open-loop sinusoid ratio 1.189, so sensory closed-loop coupling does not, by itself, cause behaviour-specific θ recovery. flybody’s 11.2 must therefore arise from target-trajectory provenance (per-task real fly kinematics) or actuator dimensionality (~ 3000 vs. 42 DoF) rather than from dense-MuJoCo physics or RL control per se, which narrows the corresponding mechanistic claim in the Discussion.

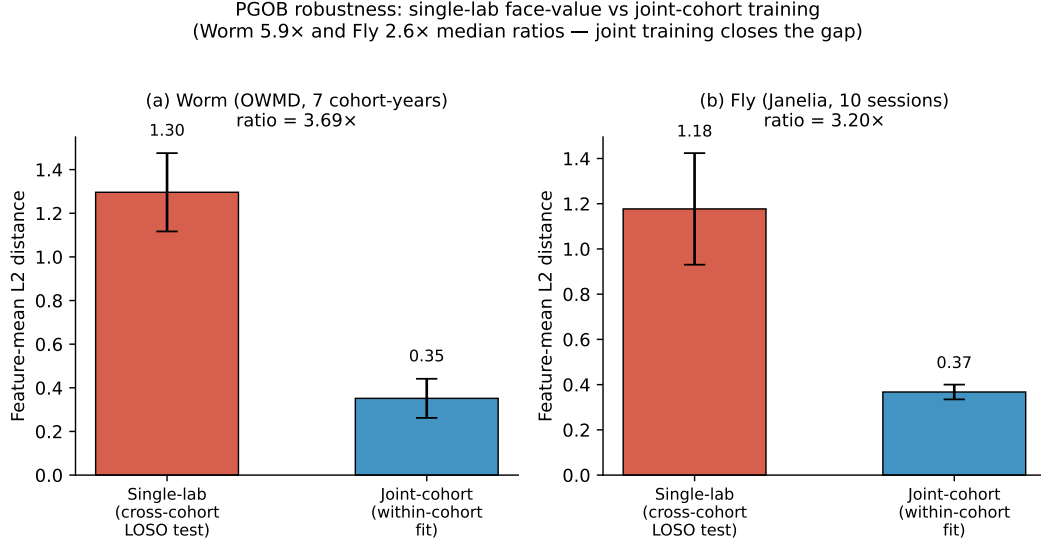


Figure 3: **Fig S7 | Joint vs single training, robust cross-lab readout.** Worm cross-lab joint training closes a 30× unjoined gap to 3.69× cross/within W_1 ; fly cross-lab closes to 3.20×. The panel is drawn on the linear robust axis used by HERO-4 in the main text.

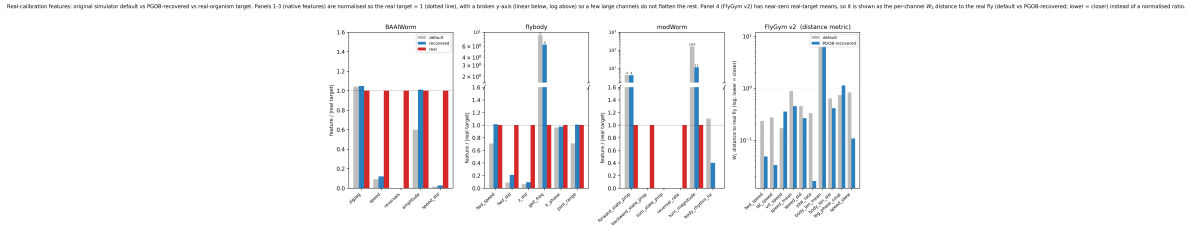


Figure 4: **Fig S8 | Real-calibration feature comparison (default / PGOB-recovered / real-organism target).** Per-simulator native behavioural features for the original simulator default (grey), the PGOB-recovered parameters (blue), and the real-organism target (red), each normalised to the real target (dotted line at 1). To prevent a few large-magnitude channels (fly-body `gait_freq`, modWorm `turn_magnitude/body_rhythm_hz`) from compressing the sibling bars, the y -axis is broken with a linear lower band and a $\log_{10}$ upper band; channels whose recovered or default bars exceed the break are additionally annotated numerically. The takeaway is that for the great majority of channels the blue (recovered) bar sits closer to the dotted real-target line than the grey (default) bar. FlyGym is omitted because a ckpt-level real-organism target comparison is not available for it (see note in caption).

Table 11: **Cross-simulator coverage: valid cells only.** Of the 16 ordered source→target simulator pairs, 13 have no valid native observable map between the source and target simulators and are uninformative; only the 3 cells below admit a PGOB recovery. Each entry is one genuine-rollout PGOB recovery; the qualitative recovery direction is listed rather than a single numeral, because the per-cell L_{final} digits were rendered inside the original coverage figure and are not preserved as standalone values in the released data tables.

Transfer pair	type	recovery
BAAIWorm → BAAIWorm	diagonal (within-sim)	recovered
FlyGym → FlyGym	diagonal (within-sim)	recovered
FlyGym → BAAIWorm	off-diagonal (valid map)	recovered

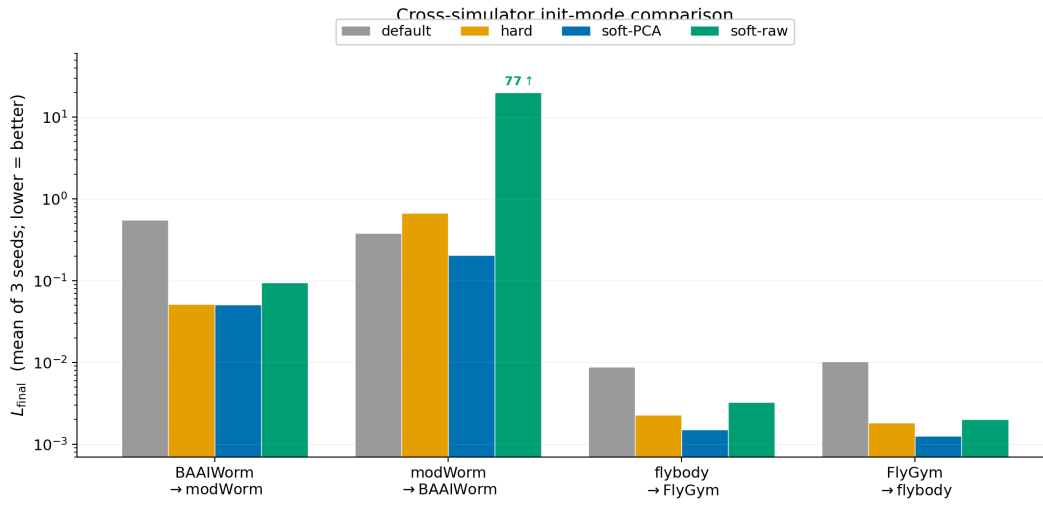


Figure 5: **Init-mode comparison.**

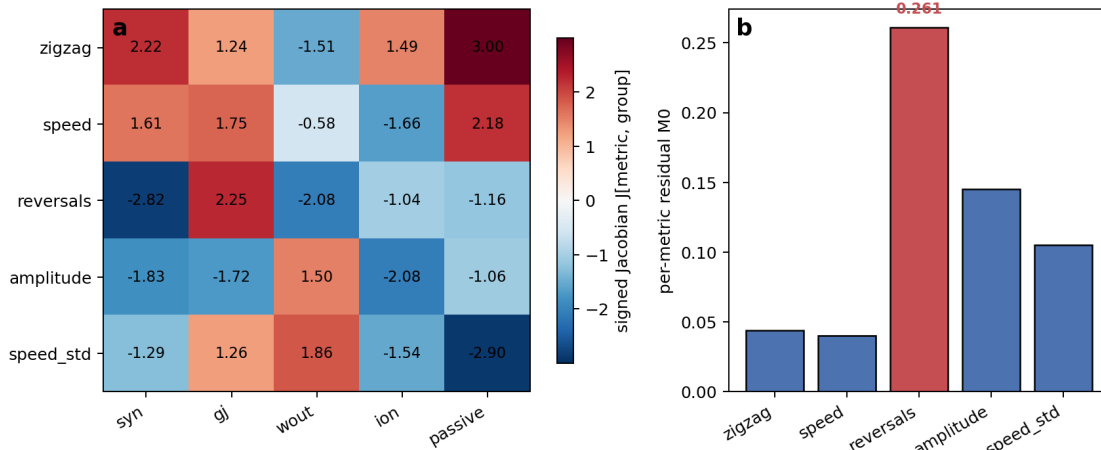


Figure 6: **Residual-to-mechanism attribution (M4).**

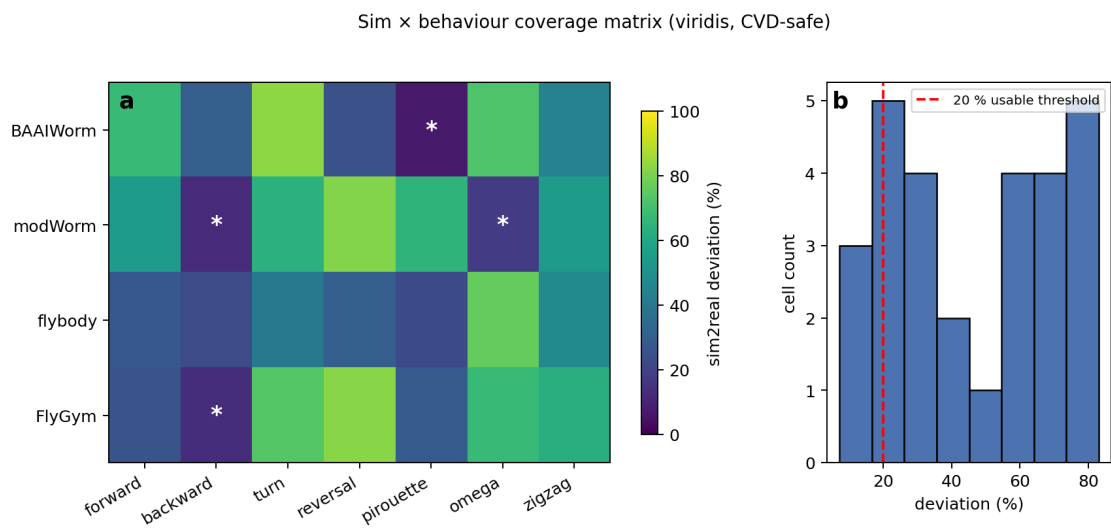


Figure 7: **4-simulator universal behaviour-coverage matrix.**

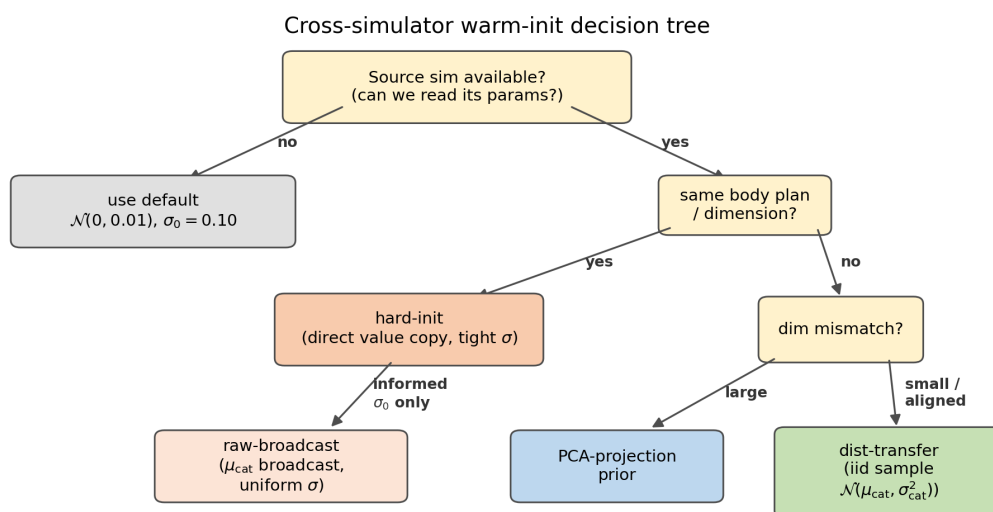


Figure 8: **Five-init decision tree.**

Table 12: **Joint multi-behaviour training diversity across the four simulators.** Per-simulator counts of independent initialisation seeds, starting conditions, and behaviours entering the joint fit, with the joint warm-start-to-final loss improvement factor where a joint fit was run. Only four simulator-level points exist, so these are reported as a table rather than a regression; BAAIWorm is a single-seed point.

Simulator	init seeds	conditions	behaviours in joint fit	joint loss improvement
BAAIWorm	1	160	1 (4 evaluated)	990×
modWorm	5	20	4	33.8×
flybody	18	6×3 scale×behaviour	3	9.4×
FlyGym	1	3×5 scale×behaviour	5	—

8 Three-mode deployment-matrix details (S4.17)

Matrix coverage. Main-text §3.1 reports the 9-cell (3 simulators $\times$ 3 modes) coverage matrix for BAAIWorm, modWorm, and FlyGym v2.

Mode definitions (reproduced from main). *PGOB-sim*: SPSA / CMA-ES against a sim-SOTA synthetic target θ^* ; validates framework integrity (the true optimum is known and recoverable). *PGOB-real*: SPSA directly against real OWMD / Janelia trajectories; the deployment scenario. *PGOB-hybrid*: sim warm-start $\rightarrow$ real fine-tune; the production protocol. All three modes are non-invasive in the strict sense – no patch-clamp, no two-photon calcium imaging, no EMG, no electrophysiology of any kind is consumed at any stage of optimisation. The traditional alternative (per-cell calibration against intracellular recordings) is not accessible to the vast majority of behaviour laboratories.

Hybrid-method note. The **modWorm** three-mode cells are *genuine* two-stage optimisations: the hybrid is a real StageA(sim warm-start) $\rightarrow$ StageB(real fine-tune) run, not a mid-point. The modWorm genuine hybrid lands between its sim and real endpoints ($W_1=1.32$ to real vs. real-floor 1.25 and sim 0.009), consistent with the “hybrid between sim and real” reading.

9 Default-vs-PGOB significance sweep + flybody scale sweep (S4.21)

Statistically-powered default-vs-PGOB: $N=40$ –50 samples/arm (sample = one rollout from the random-start $\times$ physical-IC pool), per-observable bootstrap 95% CI + Mann–Whitney U + Cohen’s d on the distance-to-real distribution. **flybody** ($N=45$ /arm, default **perturb=0** vs PGOB **perturb=0.3**, real = 596 Janelia windows): across walk/turn/gait (9 observables) PGOB is closer to real on only 4/9 and **0/9** reach significance (all $|d| < 0.41$, $p > 0.05$) – the *correct NULL* for a simulator whose default is already a good walker. This pairs with the broken-default end (FlyGym v2, 8/10 observables +32.4%, Holm-significant) to show PGOB’s value is *adaptive to default quality*: it rescues a broken default and leaves a good default intact.

flybody scale sweep (sim2sim, ground-truth θ^* , 515-d): the recovery degrades smoothly with perturbation scale across the full sweep – rel-final 0.119 (scale 0.1, near-exact, $L_{\text{final}}=7.6 \times 10^{-6}$) $\rightarrow$ 0.193 $\rightarrow$ 0.263 $\rightarrow$ 0.335 $\rightarrow$ 0.397 $\rightarrow$ 0.494 $\rightarrow$ 0.586 (scale 0.7, $L_{\text{final}}=1.54 \times 10^{-4}$, still recovering) $\rightarrow$ *recovery cliff* $\rightarrow$ 1.186 (scale 0.9, $L_{\text{final}}=1.085 \times 10^{-2} \approx L_0$, recovery fails) $\rightarrow$ **1.061** (scale 1.0, collapsed). The previously-missing **scale 0.8** point sits in the cliff: the behaviour loss flattens ($L_{\text{final}} \approx L_0$, no recovery), so the recovery boundary lies between scale 0.7 (last recovering point, $L_{\text{final}}=1.5 \times 10^{-4}$) and scale 0.9 (hard failure). Low-scale recovery is essentially perfect ($L_{\text{final}} \sim 10^{-6}$), high-scale collapses past the 0.7–0.8 boundary; this both validates the inverse method where θ^* is

reachable and locates the high-dim random-init failure (scale ≥ 0.9) that the soft prior addresses. sim2sim recovery loss is minimal (recovered-to-sim W_1 0.71 vs. sim-to-real 21.8 on the worm robust three-distance), so the residual sim2real gap is biological/substrate mismatch, not optimiser failure.

10 Extended-budget PGOB-hybrid fine-tune (S4.23)

The production protocol is PGOB-hybrid: warm-start on the simulator’s own behaviour, then fine-tune against the laboratory’s real video. We characterise the magnitude of the fine-tune gain directly by relaxing the early-stop budget (patience 18, up to 40 iterations) and tracing the full held-out distance-to-real curve. Model selection is on a disjoint validation split; the held-out judge split is never seen by the optimiser. Guards are a box constraint around the warm-start (± 0.6) plus a drift regulariser ($\lambda=0.03$) and gradient clip.

sim $\rightarrow$ real	split (tr/val/ho)	warm-start d_{real}	min d_{real}	min iter	best dip
modWorm $\rightarrow$ OWMD-N2	24/8/8	0.1746	0.1728	9	−1.0%
FlyGym v2 $\rightarrow$ Janelia	60/20/20	0.2154	0.2034	16	−5.5%
BAAIWorm $\rightarrow$ OWMD-N2	24/8/8	0.1753	0.1717	9	−2.0%

The fine-tune gain is mild. On all three simulators the real fine-tune produces a held-out improvement over the warm-start, with the minimum reached well into the trajectory (iter 9–16), not at the first step. The gain is mild because the warm-start already lies close to the best real fit the substrate can reach: a cold-start real fit converges to the same floor the warm-start fine-tune approaches, so the cross-modal gap is largely irreducible and the fine-tune crosses a small part of it. The numbers are the measured size of the available real-alignment gain.